Verification Relocates Trust: The Specification-Legibility Condition on Machine-Checked Proofs

Abstract

Suppose an opaque artificial system announces a proof of a famous open conjecture, together with a certificate that a small proof-checking kernel has verified. What exactly may a mathematician now know, and on what basis? I argue that formal verification does not eliminate the need for trust in the system's output; it relocates that trust from the proof to the *statement*. A machine-checked derivation transmits knowledge of the intended mathematical proposition to a person only to the extent that the person can independently grasp that the formal theorem is the intended proposition. I call this the **Specification-Legibility Condition** (SLC). I defend the SLC against two rivals, the *track-record view* and the *kernel-trust view*, draw out three consequences (a limit on what a certificate can do for a reader, a norm for how machine-generated results should be presented, and a diagnosis of why informal exposition retains value even when a formal proof exists), and answer objections.

1. The puzzle

Mathematics has long relied on testimony. Hardwig (1985) argued that in modern science, knowers often rationally depend on others whose reasoning they cannot reproduce, so that the epistemic subject is sometimes a community rather than an individual. Tymoczko (1979) argued that the computer-assisted proof of the four-color theorem differs from traditional proof because it cannot be surveyed by a human mind, and so seemed to turn mathematics partly empirical. Burge (1998) replied that a computer proof can yield a priori knowledge, provided the relevant steps are suitably underwritten by the reliability of the processes involved and by the epistemic entitlement we have to rely on other minds and instruments. Thurston (1994) pushed in another direction: what mathematicians want from a proof is not merely certification of a theorem but human understanding.

New developments sharpen the old questions. Large AI systems can now produce long arguments that no one has read, and some of these arguments come with formalizations in a proof assistant such as Lean, whose kernel checks that each inference follows from the axioms. The natural thought is that this finally dissolves the problem: if the kernel accepted the proof, we do not need to trust the opaque system at all, because trust in its *process* is replaced by checking of its *product*.

The thought is half right. The product is checked. But what is checked is a derivation of a *formal sentence*, and there is a further question, which the kernel does not and cannot answer: is that formal sentence the proposition we cared about? My thesis is that this question is where the residual trust sits, and that the epistemology of machine-generated mathematics should be organized around it.

2. Three things a certificate could be certifying

Let P be an informally stated mathematical proposition of interest (say, a conjecture posed in natural language and stated in the literature with conventions that experts share). A machine-checked result consists of a formal sentence φ and a derivation d such that the kernel accepts d as a derivation of φ from the foundational axioms Γ of the system. We can distinguish three claims.

1. **Derivability.** d is a derivation of φ from Γ. The kernel certifies this.
2. **Fidelity.** φ is a faithful formalization of P: φ holds in exactly the circumstances in which P holds, under the intended reading of P's terms.
3. **Foundational soundness.** Γ is consistent and true of the intended mathematical subject matter, and the kernel implementation is correct.

Of these, (1) is the target of mechanical checking. (3) is a standing background assumption shared by all of contemporary mathematics; it affects every proof, human or machine, equally, and I set it aside. Fidelity (2) is

the interesting case. It is not a mathematical claim within the formal system at all. It is a claim *about the relation between a formal object and an informal practice*, and so it cannot be proved inside the system. It can be supported only by inspection, by comparison, and by the judgment of people who understand both the formal definitions and the informal proposition.

This is not a merely theoretical worry. Formalization requires many choices: how to encode real numbers and function spaces, which regularity conditions to build into definitions, whether a quantifier ranges over what the informal text intended, whether a definition silently makes a hypothesis stronger or weaker. A formal statement can be *derivable and yet fail to be the claim of interest* in at least four ways: it can be **too weak** (it proves something easier), **too strong** (it proves something that is false of the intended objects, in which case a derivation would reveal inconsistency in Γ or, more likely, an encoding error elsewhere), **vacuous** (hypotheses are jointly unsatisfiable, so the statement holds trivially), or **off-target** (it concerns a variant of the problem with different boundary or forcing conditions). Every one of these is a failure of (2), not of (1), and the kernel is indifferent to all of them.

The interesting claim is therefore this: for a reader who wants to learn P from a machine-checked proof, the epistemic work done by the certificate is exactly the work of (1), and the work needed for the reader to come to know P includes an act of (2) that cannot be delegated to the kernel.

3. The Specification-Legibility Condition

I propose the following.

> **SLC.** A reader R acquires knowledge that P from a machine-checked derivation of φ only if R is in a position to recognize, on the basis of R's own understanding (or of testimony that R is entitled to trust for independent reasons), that φ is a faithful formalization of P.

The SLC has three features worth noting.

First, it is a *necessary* condition, not a sufficient one. A reader may satisfy it and still fail to know P for other reasons (the foundations might be unsound, the kernel buggy).

Second, it permits dependence. R need not personally inspect every definition. The condition allows that R is entitled to rely on a specialist's judgment that φ is faithful, so long as that reliance is itself grounded in something other than the very process that generated the proof. This is why the SLC is compatible with Hardwig's picture of dependence, and why it is not an individualistic demand for personal survey in Tymoczko's sense.

Third, it asks for a *separate* channel. If the only evidence that φ is faithful to P is the announcement by the system that produced φ, the condition is not met, because then the fidelity claim inherits exactly the opacity that formalization was supposed to remove.

Why accept the SLC? The argument is short.

1. To know P from a derivation of φ, R must have a justified belief that if φ then P, in the sense that the truth of φ settles the truth of P. (Otherwise the derivation of φ is evidentially irrelevant to P.)
2. That belief is not derivable in the formal system, since it concerns the relation between φ and an informal proposition.
3. So R's justification for it must come from outside the formal derivation: from R's understanding of the definitions, or from trusted testimony about them.
4. Hence R knows P from the derivation only if such an external justification exists, which is what the SLC says.

Premise 1 is the point most likely to be resisted, so it deserves a gloss. The claim is not that R must believe φ and P are logically equivalent, only that the evidential bearing of the certificate on P runs through a bridge principle linking φ to P. A reader with no basis for that bridge learns, at most, that some formal sentence has a derivation.

4. Two rivals

4.1 The track-record view

On this view, we may rationally accept a system's claim that φ faithfully formalizes P because the system has a good track record at such formalizations. This is a familiar kind of reliabilist testimonial justification. The view has some appeal: an AI system that has formalized thousands of statements correctly has earned some trust.

The reply is that track records generalize badly exactly where the stakes are highest. The reference class matters. A system's reliability at formalizing *routine* statements is weak evidence for its reliability at formalizing a *famous, subtle, and heavily contested* statement, because in famous problems the informal statement is often underdetermined in precisely the places where encoding choices matter (what counts as admissible initial data, what regularity is demanded, which variant is meant). Moreover, errors of fidelity are *silent*: a wrong formalization yields no error message. A track record is therefore only as good as the independent audits that determined it, and those audits are themselves acts of (2) performed by humans. The track-record view does not replace the SLC; it presupposes that someone, somewhere, has satisfied it for the cases that make up the record.

4.2 The kernel-trust view

On this view, what matters is that the kernel is small and well understood, so that trusting the kernel is cheaper and safer than trusting a human referee. I agree with the premise and reject the conclusion. The kernel's smallness is a virtue with respect to (1) and (3). But the kernel's smallness is *irrelevant* to (2), because (2) is not a property of the kernel's operation. The kernel-trust view mistakes the shrinking of one part of the trusted base for the shrinking of all of it. The trusted base of a machine-checked result is the kernel *plus* the statement's formalization, and while the former has been made small, the latter is as large and as subtle as the informal problem itself.

5. Consequences

5.1 A limit on what a certificate can do

The SLC implies that a certificate has *diminishing* epistemic value for a reader as the legibility of the formal statement decreases. Take two cases with identical kernel checks. In the first, the formal statement is two lines of readable definitions that a trained mathematician can compare against the informal problem in an afternoon. In the second, the statement unfolds into thousands of lines of interlocking definitions, auxiliary structures, and encoding conventions. The kernel's verdict is the same, but the reader's epistemic position is different. In the first case the reader can discharge the fidelity requirement herself. In the second she must rely on the judgment of others, and the question becomes whether those others have independent grounds.

This yields a gradient rather than a threshold, and it explains a common intuition that "a formal proof of a famous result" can be at once overwhelmingly strong evidence and curiously unsatisfying. It is overwhelming with respect to derivability and unsatisfying with respect to the claim that we learned what we wanted to learn.

5.2 A norm for presenting machine-generated results

If knowledge transmission depends on the legibility of the statement, then those who announce machine-generated results have a duty to make the bridge from φ to P as inspectable as they can. I suggest three components of such a norm:

- **Statement isolation.** The final theorem statement should be separable from the proof infrastructure, so that a reader can read the statement and its dependencies without reading the proof.
- **Definition audit trail.** Each non-standard definition on which the statement depends should be linked to the informal definition it is meant to capture, with an explanation of encoding choices that could affect the result.
- **Variant disclosure.** Where the informal problem admits several variants (different boundary conditions,

with or without forcing, different regularity classes), the announcement should say which variant the formal statement addresses, and should not describe the result by the name of the broader problem without qualification.

None of these demands is specific to machines. But machine production changes the *default*. Human-produced proofs typically come with an author who can answer questions about what was meant. An opaque system cannot be questioned in the same way, and its human custodians may not themselves know every encoding decision. For that reason the burden of legibility shifts toward the presentation.

5.3 Why informal exposition retains value

Thurston's observation that mathematicians seek understanding takes a new form under the SLC. An informal exposition of the proof is not just a convenience for human readers; it is often the best available evidence for fidelity. If a proof's informal outline uses the same objects and conditions as the informal problem, and the formal statement is visibly the formalization of *that* outline, then the exposition ties the formal result to the intended proposition. Conversely, an exposition that diverges from what was formalized is a warning sign. This gives a principled reason to want both artifacts and to check that they agree, and it explains why discrepancies between a written argument and a verified formalization, even when each is internally fine, matter to the community.

6. Objections and replies

Objection 1: The SLC proves too much. Human proofs also depend on a fidelity step, namely the author's claim to have proved what the problem asks.

Reply: Agreed, and this is a feature. In human mathematics the fidelity step is cheap because the proof and statement are in the same natural language, so that anyone who can read the proof can see that it addresses the stated problem. Formalization separates the two languages and thereby introduces the gap explicitly. The SLC does not say that machine-checked proofs are worse than human ones, only that the place at which trust is needed has moved.

Objection 2: Fidelity can be checked by running tests, or by proving sanity-check lemmas about φ, and so can be brought inside the formal system after all.

Reply: Such checks can establish *consistency conditions* (for instance, that the hypotheses are satisfiable, or that φ implies a known special case). They are valuable, and they address the "vacuous" and "too strong" failure modes. They cannot establish that φ is the *intended* statement, because intention is not a formal object. Sanity lemmas shrink the space of possible fidelity failures but do not close it. A further sanity lemma is itself a formal sentence whose fidelity to *its* informal purpose must be assessed, so the regress, though it can be productive, is not eliminable.

Objection 3: In practice, communities of specialists will audit statements, so the SLC is satisfied by testimony and the philosophical point is idle.

Reply: This is exactly the SLC working as intended. The condition says that such external grounding is *required*, and the practical question is how well it works. The philosophical content is that it is required, and that the grounding must not itself reduce to the opaque system's say-so. The point is not idle, because it tells us what to demand when an announcement arrives: not only a certificate, but a legible statement and an independent audit.

Objection 4: If the opaque system produces both the formalization and the proof, and the formalization is correct, then the reader has knowledge even without checking, since reliable processes yield knowledge regardless of the reader's grounds.

Reply: This is an externalist objection. On a crude reliabilism, a reader could know P simply because the process that produced her belief is in fact reliable. I do not deny that such a reader may *have a true belief produced by a reliable process*. My claim concerns the conditions under which a reader's belief is *epistemically responsible* in a context where its sources are opaque and where fidelity failures are silent. In

such contexts a reliabilist should also ask about the reliability of the *method of acceptance*, and a method of accepting any certified result whatever its statement's legibility is not reliable across the range of statements it would accept. So even the externalist should endorse a practical analogue of the SLC.

Objection 5: The SLC makes the epistemic situation of non-experts hopeless, since most people cannot read formal statements.

Reply: Non-experts rely on testimony from experts about nearly everything in mathematics. The SLC allows this reliance, provided that the testimony is independent of the generating process. What the SLC rules out is deference to an announcement by the very party that benefits from the claim being accepted. The remedy is institutional: independent audits, and norms of presentation as in §5.2.

7. A worked contrast

To see the condition operate, compare two hypothetical announcements.

Announcement A. A system outputs a derivation of a formal sentence. The accompanying note says: "This resolves Conjecture C." The formal sentence is hundreds of lines long, relies on a custom library of several thousand lines, and the note does not say which variant of C is addressed.

Announcement B. A system outputs a derivation of a formal sentence. The accompanying note says: "This resolves the version of Conjecture C in which hypothesis H holds; the formal statement is lines 1–40 of the file, which define the relevant objects in terms of standard library notions; the one nonstandard definition is explained in section 2; the variant in which H is dropped remains open." Two independent groups have confirmed the correspondence between the statement and the informal problem.

The kernel's verdict is identical in both. By the SLC, a reader is much better placed to know the resolved proposition in B than in A, though in neither case does she need to follow the proof. And a reader of A who is told only that "the kernel accepted it" has been told something true and almost beside the point.

8. Conclusion

Formal verification is one of the great achievements of the foundations of mathematics, and it can reduce the role of trust in the *reasoning* to a very small kernel. But it cannot reduce to zero the role of trust in the *meaning* of what is proved, because meaning is the relation between formal object and informal practice. The Specification-Legibility Condition states the consequence: knowledge from a machine-checked proof requires that the fidelity of the statement be recognizable by the reader, or by testimony she may independently trust. This explains the gradient of epistemic value across statements of differing legibility, recommends concrete norms for presenting machine-generated results (statement isolation, definition audit, variant disclosure), and shows why informal exposition remains evidentially important even in a world of certificates. As machine-generated proofs of significant results become more common, the philosophically pressing question will not be whether the kernel can be trusted but whether anyone besides the generating system can say what the theorem says.

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